VSDBabySoC

POST-ROUTE TIMING AND PHYSICAL-DESIGN STUDY

Sky130HD standard cells | OpenROAD flow | OpenSTA analysis

Report purpose

This document interprets the available post-route timing evidence, explains how physical implementation affects timing, and records the remaining analysis and layout issues. It is organized around the design flow, the measured corner results, and the engineering actions needed next.

Evidence boundary

The documented data covers three exploratory single-library runs only: TT_025C_1v80, SS_n40C_1v28, and FF_n40C_1v95. This is not a complete 16-corner analysis or a sign-off report.

At-a-glance results

Run	Setup slack	Hold slack	Current reading
TT_025C_1v80	+5.99 ns	+0.31 ns	Reported MET
SS_n40C_1v28	-30.73 ns	+1.76 ns	Setup violation
FF_n40C_1v95	+7.99 ns	+0.19 ns	Reported MET

Physical-design warning: the saved route DRC report records 14 short violations. Therefore, the layout is not DRC-clean on the available evidence. The SS setup violation and other constraint/model limitations also remain open.

Prepared from the supplied report and embedded flow screenshots. Added explanatory sections provide technical context; they do not represent additional measured results.

Report roadmap

The report follows the implementation process first, then explains the timing model and the results, and finishes with limitations and a corrective plan. This ordering connects the screenshots to the checks they help illustrate.

Part	What it covers
1. Flow overview	Synthesis through routing and extraction; purpose of each stage.
2. Timing fundamentals	Setup, hold, slack, WNS/TNS, PVT corners, and parasitics.
3. Analysis inputs	Database, SPEF, SDC, Liberty, and clock configuration.
4. Physical-design evidence	Reordered screenshots with stage-specific captions.

5. Measured results	Corner-by-corner interpretation and the unresolved TT discrepancy.
6. Constraint/model audit	I/O delays, unconstrained endpoints, and macro timing limitations.
7. DRC review	Meaning of the 14 reported shorts and separation from timing closure.
8. Debug strategy	Recommended sequence for reproducing and fixing the issues.
9. Conclusion and appendix	Current status, terminology, and referenced artifacts.

1. From RTL to post-route timing

A digital implementation flow progressively transforms a logical design into a physical layout. Each stage produces artifacts that become inputs to later stages. Timing conclusions should always be tied to the stage and models used to generate the report.

Stage	Purpose	Evidence in this report
Synthesis	Maps the logic into standard-cell instances using the selected li brary.	Figure 1
Floorplanning	Defines the core area, rows, and initial physical arrangement.	Figure 2
Placement	Assigns locations to standard cells; placement influences wirel engths and congestion.	Figure 3
Clock-tree synthesis	Builds the clock distribution network and introduces clock-tree delays and skew.	Figure 4
Routing	Creates physical interconnect on routing layers.	Figures 5–6
Parasitic extraction + S TA	Uses extracted wire RC and timing models to evaluate constrai nts.	Figures 7–8 and corner reports

The supplied screenshots visually document several flow stages, but images alone do not establish that every stage completed cleanly. The saved reports—not the appearance of the layout window—are the basis for the timing and DRC conclusions in this document.

2. Timing fundamentals for post-route analysis

2.1 Setup and hold checks

A setup check asks whether data reaches a capture register early enough before its active clock edge. A hold check asks whether the data remains stable long enough after that edge. These checks evaluate different failure mechanisms: a long data path can fail setup, while an excessively short path can fail hold.

For the reported checks, positive slack means the check has margin and negative slack means it violates the required timing. A positive hold result does not compensate for a negative setup result; both must be considered independently.

2.2 Slack, WNS, and TNS

Slack is calculated as required time minus arrival time for the relevant check. Worst Negative Slack (WNS) identifies the most negative setup slack among the paths included in the report. Total Negative Slack (TNS) aggregates negative slack across violating paths. TNS depends on the report's endpoint/path scope and constraints, so values should only be compared when report conditions are consistent.

2.3 Why post-route timing can differ from earlier estimates

Before routing, interconnect delay is estimated from less detailed physical information. After routing, the design can be analyzed with extracted parasitic resistance and capacitance from the routed wires. Longer wires, coupling assumptions, fanout, buffering, and clock-tree behavior can change delay and slew. Consequently, pre-route and post-route slack are not interchangeable.

2.4 PVT corner meaning

PVT stands for process, voltage, and temperature. The TT, SS, and FF labels identify typical, slow, and fast process-model families; the suffixes identify the temperature and supply voltage used by the selected library. A corner is a model

condition, not a guarantee that every path behaves uniformly. The critical corner depends on the path type, cell arcs, interconnect, and clock relationships.

2.5 Clock and interface constraints

The clock period sets a central timing requirement for synchronous paths. Clock propagation and uncertainty also affect required time. Input delays describe when external data can arrive relative to a reference clock; output delays describe the timing budget required by downstream logic. Missing interface constraints can leave paths unchecked or make reports incomplete even when the reported internal paths have positive slack.

3. Analysis configuration and input artifacts

The recorded standalone audit references the following files and timing models:

- **6_final.odb** — routed OpenDB database used for physical connectivity and layout context.
- **6_final.spef** — extracted parasitic information for interconnect delay estimation.
- **5_route.sdc** — constraints used for route-stage timing analysis.
- **Sky130HD Liberty** — standard-cell timing and electrical characterization for the selected library corner.

The reviewed SDC defines an 11.0000 ns clock at `pll/CLK` and uses propagated clock timing. The reviewed file does not define input delays or output delays.

3.1 Reproducibility checklist

To make corner-to-corner comparisons meaningful, the analysis should record the exact database, parasitic file, Liberty file(s), SDC, tool version, and report commands for each run. If any of these differ, the resulting slack values may not be directly comparable.

Item to record	Why it matters
Library and PVT corner	Determines the cell timing arcs used in calculations.
SDC and clock definition	Sets required times and determines which paths are constrained.
SPEF / parasitic source	Controls the modeled interconnect contribution to delay.
Database revision	Ensures the timing result matches the intended routed layout.
Report command and path scope	Defines which checks and paths contribute to WNS/TNS.
Tool/version and run log	Helps reproduce the exact result and diagnose discrepancies.

Open item: a saved setup report records +7.34 ns slack, while the standalone TT audit records +5.99 ns. The available evidence does not establish the reason. Keep both values labeled by source until the run conditions and report commands are reconciled.

4. Physical implementation evidence

The screenshots below are placed beside the stage descriptions they illustrate. They provide visual context for the flow and are not substitutes for numerical timing or DRC reports.

4.1 Synthesis and floorplanning

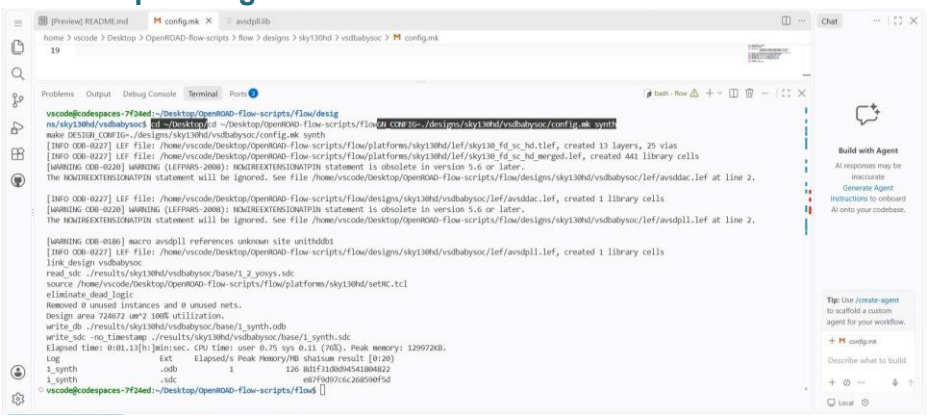


Figure 1. Synthesis-stage console evidence. The view is associated with logic synthesis and flow setup; it does not by itself demonstrate post-route timing closure.

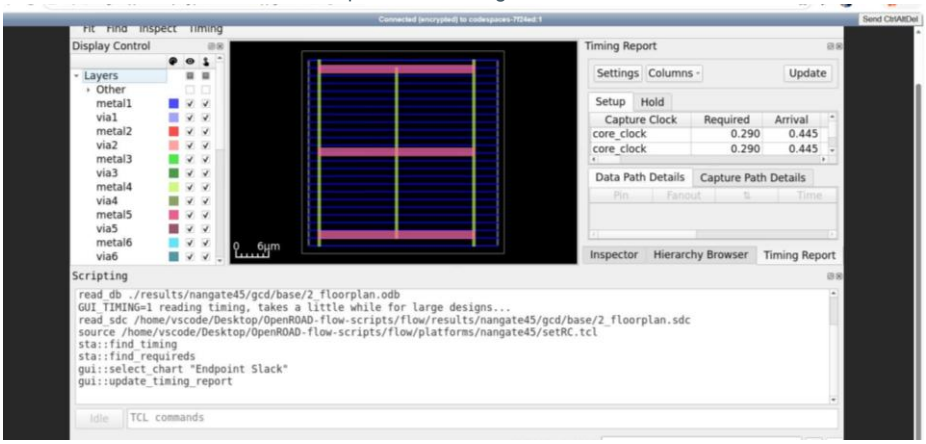


Figure 2. Floorplan-stage view. The layout canvas and timing panel provide context for early physical planning; later placement and routing can change interconnect delay.

4.2 Placement and clock distribution

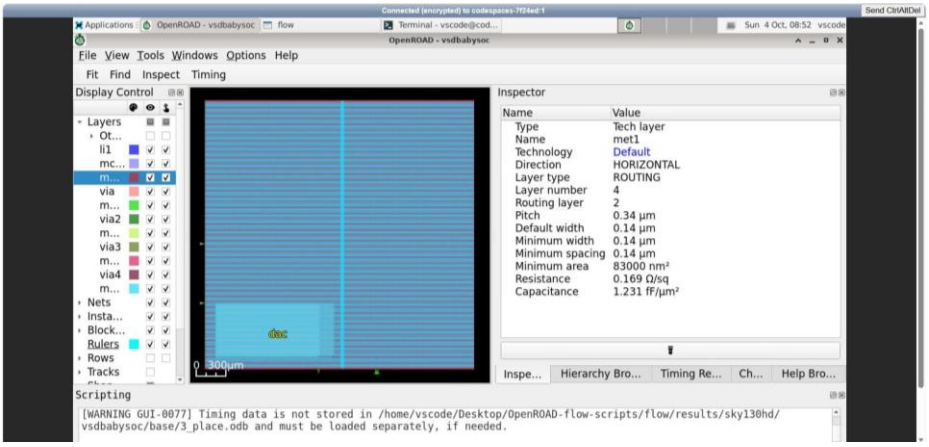


Figure 3. Placement-stage view. Cell placement affects wirelength and congestion, both of which can influence post-route parasitics.



Figure 4. Clock-tree synthesis view. Clock distribution affects insertion delay and skew; no numerical CTS quality claim is inferred from this image alone.

4.3 Routing and interconnect

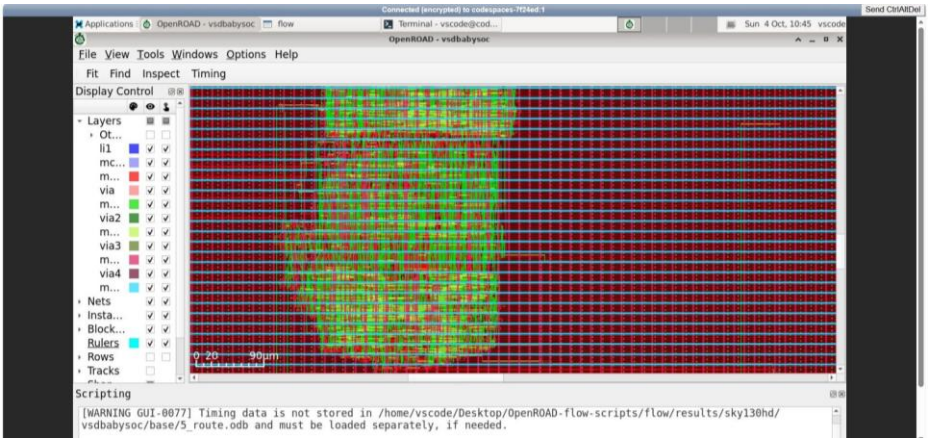


Figure 5. Routing-stage overview. The dense routing display illustrates the interconnect fabric, but visual completion is not evidence that all design rules pass.

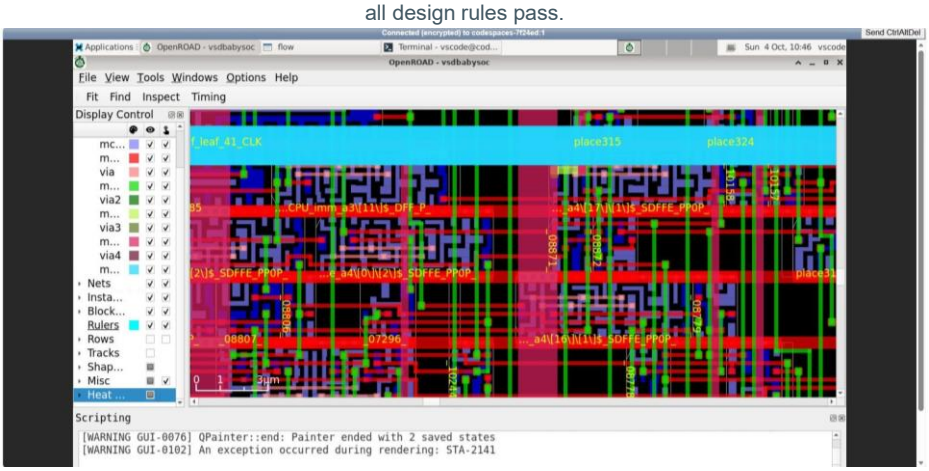


Figure 6. Detailed routing view. The closer view shows local interconnect detail. The saved DRC report nevertheless lists 14 short violations that remain to be investigated.

4.4 Static timing evidence



Figure 7. OpenSTA setup timing evidence. Use the associated corner report to identify startpoint, endpoint, path delay, required time, and slack for a reproducible interpretation.



Figure 8. OpenSTA hold timing evidence. The summary values supplied for all three documented runs are positive; full sign-off still requires complete constraints and required corner coverage.

5. Recorded corner results and interpretation

The table reproduces the measured values supplied in the source report. “Not specified” means that a corresponding value was not supplied for that corner in the summary; no missing WNS/TNS value has been inferred.

Corner	Condition	Setup slack	WNS	TNS	Hold slack
TT_025C_1v80	Typical, 25 C, 1.80 V	+5.99 ns	Not specified	Not specified	+0.31 ns
SS_n40C_1v28	Slow, -40 C, 1.28 V	-30.73 ns	-30.73 ns	-13550.81 ns	+1.76 ns
FF_n40C_1v95	Fast, -40 C, 1.95 V	+7.99 ns	Not specified	Not specified	+0.19 ns

5.1 Typical run — TT_025C_1v80

The standalone audit reports +5.99 ns setup slack and +0.31 ns hold slack, with both checks marked MET. A separate saved setup report reports +7.34 ns. Since the discrepancy has not been explained, neither value should silently replace the other. First compare the exact libraries, constraints, database/parasitics, and report commands.

5.2 Slow run — SS_n40C_1v28

The reported setup slack and WNS are both -30.73 ns, and the reported TNS is -13550.81 ns. This indicates a substantial setup failure in that run and suggests that the issue affects more than a single isolated path. The available summary does not identify the root cause; path-level reports are needed to separate cell delay, net delay, clock effects, and constraint effects.

The reported hold slack is +1.76 ns. This means the reported hold check passed for that run, but it does not make the setup result acceptable.

5.3 Fast run — FF_n40C_1v95

The standalone report gives +7.99 ns setup slack and +0.19 ns hold slack, with both checks marked MET for the reported run. This result is encouraging for that particular library condition, but it cannot be generalized to all corners or used to dismiss the SS violation.

5.4 What the results do and do not establish

- They establish the reported slack values for three individual runs.
- They show a setup violation in SS_n40C_1v28 and positive reported hold slack in all three runs.
- They do not establish complete 16-corner coverage, full MCMM sign-off, or complete interface/macro timing coverage.
- They do not prove the layout is DRC-clean; the route DRC report lists 14 shorts.

6. Constraint coverage and macro-model limitations

The standalone audit lists the following constraint and model issues:

- Six input ports lack input delays: `ENb_CP`, `ENb_VCO`, `REF`, `VCO_IN`, `VREFH`, and reset.
- Output port `OUT` lacks an output delay.
- Two endpoints were reported as unconstrained.
- The available PLL and DAC Liberty files (`avsdpll.lib` and `avsddac.lib`) contain no timing arcs, limiting the timing behavior modeled for those macros.

These gaps mean the current reports may not cover every external interface requirement or macro-related timing behavior. Interface delays and loads should be based on the actual system specification, not guessed. The PLL/DAC models should be reviewed to determine whether complete timing models or documented assumptions are appropriate.

6.1 Why unconstrained paths matter

A path that lacks the necessary timing constraints may not be evaluated against the intended system requirement. A report with good slack on constrained paths can therefore still be incomplete. Before treating results as sign-off evidence, confirm that all intended clocks, input paths, output paths, exceptions, and macro interfaces are covered.

7. Design-rule status and timing closure

The saved `5_route_drc.rpt` records 14 short violations, mainly involving met1 and including DAC-related instances. On the evidence available, the route is not DRC-clean. The detailed DRC markers should be inspected to identify the affected nets, layers, and instances; after correction, routing and DRC must be rerun.

Timing closure	Physical-rule closure
Checks data arrival against setup/hold requirements under clock constraints.	Checks geometric and connectivity rules in the physical layout.
Typical evidence: setup/hold reports, slack, WNS/TNS, path details.	Typical evidence: detailed DRC markers and a clean final DRC report.
Can fail even if the layout has no DRC errors.	Can fail even if selected timing checks report positive slack.

The two closure activities are related but not interchangeable. Neither a positive slack value nor a visually routed layout is sufficient to claim complete sign-off.

8. Recommended debug and validation plan

The following sequence is intended to make the next investigation reproducible and prevent one unresolved issue from being mistaken for a completed fix.

Step 1 — Freeze and identify the baseline

Record the exact routed database, SPEF, SDC, Liberty files, OpenROAD/OpenSTA versions, run command, and report output for each corner. Keep each run's output separate and label the corner explicitly.

Step 2 — Reconcile the TT discrepancy

Re-run the TT analysis using identical inputs and report commands. Compare the source of +7.34 ns and +5.99 ns, including whether the database, parasitics, constraints, library, and path scope are identical. Document the reason only after it is demonstrated.

Step 3 — Inspect the worst SS setup paths

Generate detailed max-delay reports for the worst violating paths. For each path, capture startpoint, endpoint, cell sequence, cell delay, net delay, arrival time, required time, clock path, and uncertainty. Classify the dominant contributors before selecting a fix.

Step 4 — Validate constraint completeness

Confirm the intended clock and interface timing from the design specification. Add input/output delays and electrical constraints only where justified. Investigate the two unconstrained endpoints and verify the intended treatment of PLL/DAC timing.

Step 5 — Investigate DRC shorts

Open the detailed route DRC report and map each violation to its layer, net, and instance. Apply a physical-design correction, rerun the required routing steps, and generate a fresh DRC report. Do not carry the previous violation count forward as though it were a post-fix result.

Step 6 — Repeat the required corner/mode matrix

Once the analysis setup is stable, run the required PVT corners and operating modes with consistent constraints and report settings. Summarize setup and hold results, unconstrained paths, macro coverage, and DRC status together so that the final status reflects all outstanding checks.

Suggested closure checklist

- TT discrepancy explained and reproducible.
- SS setup violation investigated and any fix re-measured.
- Input/output constraints reviewed against the specification.
- Unconstrained endpoints understood and intentionally handled.
- PLL/DAC timing coverage reviewed.

- All reported DRC shorts corrected and final DRC rerun.
- Required corner/mode coverage completed with consistent reporting.

9. Conclusion and current status

The recorded post-route timing data shows positive setup and hold slack in the TT and FF exploratory runs. The SS_n40C_1v28 run has -30.73 ns setup slack, WNS of -30.73 ns, and TNS of -13550.81 ns, while its reported hold slack is +1.76 ns. The saved route DRC report lists 14 short violations, and the constraint audit identifies missing I/O delays, two unconstrained endpoints, and limited PLL/DAC macro timing coverage.

The design should therefore be described as being under preliminary post-route analysis, with timing and physical-design issues still open. The next priorities are to reconcile the TT setup discrepancy, investigate the SS critical paths, complete specification-based constraints, improve macro timing coverage, clear the DRC shorts, and repeat analysis across the required corner/mode matrix.

Current status: preliminary analysis; three corner runs documented; SS setup failure and DRC violations remain unresolved; sign-off readiness has not been established.

Appendix A — Terms used in this report

Term	Meaning in this report
STA	Static Timing Analysis; evaluates timing paths against constraints.
PVT	Process, voltage, and temperature condition of a timing library.
SDC	Synopsys Design Constraints file containing clocks and timing requirements.
ODB	OpenDB design database used by the physical-design flow.
SPEF	Standard Parasitic Exchange Format; stores extracted interconnect parasitics.
Liberty	Cell timing and electrical characterization format.
WNS / TNS	Worst Negative Slack / Total Negative Slack for the report's defined scope.
DRC	Design Rule Check; checks physical layout rules such as shorts and spacing.
MCMM	Multi-corner/multi-mode analysis across required library conditions and operating modes.

Appendix B — Referenced evidence artifacts

- `reports/corners/TT_025C_1v80.rpt`
- `reports/corners/SS_n40C_1v28.rpt`
- `reports/corners/FF_n40C_1v95.rpt`
- `reports/post_route_setup.rpt`
- `reports/post_route_hold.rpt`
- `reports/5_route_drc.rpt`
- `logs/post_route_audit_tt.log`
- `scripts/post_route_audit.tcl`

Source note: measured values, named artifacts, and unresolved issues are retained from the supplied report. Added sections explain standard STA concepts and propose follow-up work; they are not additional measurements or claims that a fix has succeeded.